PH.30102 **FALL 2026**

Quantum Mechanics III

Homework 13

Seonwoo Park Due 2026-11-29 23:59

Topic

Quantum Process Tomography

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Quantum Process Tomography. State all assumptions and conventions.
$$p_{jk} = Tr[(\rho_j^T \otimes E_k)J_{\mathcal{E}}]$$
2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.
$$J_{\mathcal{E}} \geq 0, Tr_{out}J_{\mathcal{E}} = I$$
3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.
$$QDC_{new} = QDC_{26} + \Delta J$$

Show enough intermediate work that another student can reproduce the calculation.